# Statistics 154 / 254: Statistical machine learning

**UC Berkeley, Fall 2026**

## Instructors



**Instructor:** Ryan Giordano Office: 389 Evans Hall Office hours: Tues + Wed 10:10am-11am Gateway-LL-Golden Quarter B1040 `rgiordano@berkeley.edu` pronouns: He / him



**GSI**: Lucas Schwengber Office: Gateway desk 3330-05 Office hours: 9-11 AM Thu/Fri Gateway-LL-Classroom B1008 (29) `lucas.schwengber@berkeley.edu` pronouns: He / him



**GSI**: Josh Davis Office: TBD Office hours: 4-6 PM Wed/Friday W: Gateway-LL-Office Hours 1040F F: Gateway-LL-Golden Quarter B1040 `joshdavis@berkeley.edu` pronouns: He / him

> **Note**
>
> Note that (due to HVAC problems) lectures will now be held in Lewis 100, starting immediately, for the foreseeable future.

This website will contain lecture materials and assignments. Day-to-day announcements and discussion will be found in ED. (See links above.)

## Materials

The course will be based largely on Leture notes. The notes can be supplemented using subsets of the following texts:

- Patterns, predictions, and actions: A story about machine learning Hardt, Recht
- ESL: The Elements of Statistical Learning Hastie, Tibshirani, Friedman
- ISL: An Introduction to Statistical Learning James, Witten, Hastie, Tibshirani

Additional reading will supplement these texts as necessary.

Some other good sources of reading material are:

- Understanding Machine Learning: From Theory to Algorithms Shalev-Shwartz, Ben-David
- Probabilistic Machine Learning: An Introduction Murphy
- Dive into Deep Learning Many authors

## Schedule

Lectures will be held Aug 27th 2026 – Dec 3rd 2026 from 3:30pm – 5pm (starting at Berkeley time, 3:40pm) in **Lewis 100**.

Labs will be held on Mondays from

- 9am–11am (Gateway B1026)
- 4pm–6pm (Gateway B1012)
- 11am–1pm (Gateway B1023)
- 1pm–3pm (Gateway B1023)

The official course catalog entry is the ground truth for scheduling. If the catalog conflicts with this webage, please notify the instructor and trust the catalog.

The tentative week-by-week course calendar is as follows. **The following schedule is aspirational and subject to change as we go.**

According to the official schedule, the final exam has not been scheduled yet .

Table 1: Calendar (tentative)

| Date | Day | Note | Unit | Topic | Assignment |
|---|---|---|---|---|---|
| Aug 27 | Thursday | Lecture | Course policies | | |
| Aug 31 | Monday | Lab | | | Python review |
| Sep 1 | Tuesday | Lecture | Unit 0: Introduction and review | | |
| Sep 3 | Thursday | Lecture | | | HW 0 |
| Sep 7 | Monday | Administrative holiday | | | |
| Sep 8 | Tuesday | Lecture | Unit 1: Regression | | |
| Sep 10 | Thursday | Lecture | | | Quiz 0 |
| Sep 14 | Monday | Lab | | | Challenge 1 |
| Sep 15 | Tuesday | Lecture | | | |
| Sep 16 | Wednesday | Add / drop deadline | | | |

| Date | Day | Note | Unit | Topic | Assignment |
|---|---|---|---|---|---|
| Sep 17 | Thursday | Lecture | | | |
| Sep 21 | Monday | Lab | | | Challenge 1 |
| Sep 22 | Tuesday | Lecture | Unit 2: Classification | | |
| Sep 24 | Thursday | Lecture | | | HW 1 |
| Sep 28 | Monday | Lab | | | Oral Exam 1 |
| Sep 29 | Tuesday | Lecture | | | |
| Oct 1 | Thursday | Lecture | | | Quiz 1 |
| Oct 5 | Monday | Lab | | | Challenge 2 |
| Oct 6 | Tuesday | Lecture | Unit 3: Risk and Complexity | | |
| Oct 8 | Thursday | Lecture | | | HW 2 |
| Oct 12 | Monday | Lab | | | Challenge 2 |
| Oct 13 | Tuesday | Lecture | | | |
| Oct 15 | Thursday | Lecture | | | Quiz 2 |
| Oct 19 | Monday | Lab | | | Oral Exam 2 |
| Oct 20 | Tuesday | Lecture | | | |
| Oct 22 | Thursday | Lecture | | | |
| Oct 26 | Monday | Lab | | | Challenge 3 |
| Oct 27 | Tuesday | Lecture | Unit 4: Trees and weak learners | | |
| Oct 29 | Thursday | Lecture | | | HW 3 |
| Nov 2 | Monday | Lab | | | Challenge 3 |
| Nov 3 | Tuesday | Lecture | | | |
| Nov 5 | Thursday | Lecture | | | Quiz 3 |

| Date | Day | Note | Unit | Topic | Assignment |
|---|---|---|---|---|---|
| Nov 9 | Monday | Lab | | | Oral Exam 3 |
| Nov 10 | Tuesday | Lecture | | | |
| Nov 11 | Wednesday | Administrative holiday | | | |
| Nov 12 | Thursday | Lecture | | | |
| Nov 16 | Monday | Lab | | | Challenge 4 |
| Nov 17 | Tuesday | Lecture | Unit 5: Neural networks and optimization | | |
| Nov 19 | Thursday | Lecture | | | HW 4 |
| Nov 23 | Monday | Lab | | | Challenge 4 |
| Nov 24 | Tuesday | Lecture | | | |
| Nov 25 | Wednesday | Thanksgiving | | | |
| Nov 26 | Thursday | Thanksgiving | | | |
| Nov 27 | Friday | Thanksgiving | | | |
| Nov 30 | Monday | Lab | | | Oral Exam 4 |
| Dec 1 | Tuesday | Lecture | | | Quiz 4 |
| Dec 3 | Thursday | Lecture | Class review | | HW 5 |